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Karl Pearson and the Origin of Kurtosis

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Summary

Although the kurtosis index proposed by Karl Pearson in 1905 is introduced in statistical textbooks at all levels, the measure is not easily interpreted and has been a subject of considerable debate. In this study, the theoretical development of kurtosis is surveyed from a historical perspective of Pearson's work on evolution. It surprisingly emerges that there was no emphasis in Pearson's papers on kurtosis as measuring (in part) tail heaviness. However, it is found that Pearson used to frequently adjust the formalisation of kurtosis depending on his changing needs. This complex development partly explains the confusion that would surround kurtosis in subsequent literature. Our conclusion is that most misunderstandings arise from improper use of the kurtosis coefficient outside the Pearson system of frequency curves.

Key words: Standard fourth moment; Excess; Flat-toppedness; Tailedness; Pearson system.

1 Introduction

During the late 19th century, it was an extensive practice among statisticians to treat any frequency distribution as normal. Histogram data displaying plurimodality were typically fitted with normal mixtures, skewness was removed at the outset by transformations to normality, and symmetry was regarded as an indisputable evidence of a Gaussian population.

As an alternative, Karl Pearson (1857–1936) proposed a system of theoretical curves with a wide variety of shapes so as to provide more accurate representations of observed data. The classification of distributions inside the Pearson system was entirely determined by the two moment ratios

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}, \quad \beta_2 = \frac{\mu_4}{\mu_2^2} \tag{1}$$

associated with the concepts of *skewness* and *kurtosis*, respectively. Scale-invariant by construction, these quantities performed well as shape factors, enabling Pearson to contest, in many practical situations, the normality assumption made by his contemporaries. Still, although the relationship of β_1 to skewness was rather "transparent" and intuitive, major difficulties arose with the interpretation of β_2.

For this reason, statistical textbooks have traditionally described kurtosis in one short paragraph, following Yule & Kendall's (1937) historical definition: "The coefficient β_2 is used to measure a property of the single-humped distribution known as kurtosis ($\kappa\upsilon\rho\tau o\varsigma$, 'humped). We take as standard value of β_2 the number 3, for reasons which will appear when we study the

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so-called "normal" curve (...) Curves with values of β_2 less than 3 will, compared with this, be flat-topped, and are called platykurtic ($\pi\lambda\alpha\tau\upsilon\varsigma$, 'broad, + $\kappa\upsilon\rho\tau o\varsigma$). Curves with values greater than 3 will be peaked more sharply, and are called leptokurtic ($\lambda\varepsilon\pi\tau o\varsigma$, 'narrow', + $\kappa\upsilon\rho\tau o\varsigma$)".

A mnemonic trick attributed to "Student" seeks to clarify the relationship of kurtosis to shape with the help of two animals: "Platykurtic curves, like the (duck-billed) *platypus*, are squat with short tails. Leptokurtic curves are high with long tails, like the kangaroo, noted for *lepping*" (Dyson, 1943).

In order to reconstruct the exact origin of these definitions, we survey here the theoretical development of kurtosis through a series of papers published by Karl Pearson between 1894 and 1905. Starting with a vague notion of *excess* of a symmetrical frequency curve relative to the normal (1894), Pearson arrived at the *characteristic coordinate* β_2 (1895–1901); this evolved into the *degree of flat-toppedness* (1902) and eventually became the *kurtosis* coefficient that is in common use today (1905).

The history of kurtosis prior to 1905 is hardly mentioned in the literature, yet it provides a useful insight into Pearson's way of thinking and approaching statistics. We focus on the repeated changes in Pearson's formalisation of kurtosis depending on his changing needs. At first, the index of excess provided statistical evidence against contemporary attitudes that regarded all symmetrical distributions as normal. Then, the concept of kurtosis helped Pearson face criticism against his newborn system of frequency curves. In this complex development, it surprisingly emerges that there was no emphasis in Pearson's work on kurtosis as measuring (in part) tail heaviness, which seems to be its more frequent contemporary usage. In fact, the degree of kurtosis was designed to classify symmetrical curves of the Pearson system exclusively according to their peakedness at the mean.

The rest of the paper is organised into five sections. In the first four, we elaborate on Pearson's articles involving major steps in the derivation of his kurtosis measure. Our conclusions are summarised in the last section. We maintain the notation originally adopted by Pearson and use quotation marks when reporting his original statements.

2 1894: From Variations in Crustaceans to the Index of Excess

The origins of kurtosis are to be found in the friendship of Karl Pearson and Walter F. R. Weldon at University College, London.

Their first contact took place in 1891, a few years after Pearson's appointment to the Chair of Applied Mathematics (1885). By that time, Weldon was a young zoologist fascinated by the idea of studying animal evolution through morphological measurements. Inspired by Galton's *Natural Inheritance* (1889), he aimed at extending the domain of the normal curve to variations in crustaceans. It was quite natural for that time to relate a normal-like behaviour in biological measurements to a population approaching a stable condition ("There is production and destruction impartially round the mean": Pearson, 1894). However, Weldon's collection of data on shrimps and shore crabs exhibited well-marked departures from normality, including both skewness and abnormal modal frequencies. This raised the question: Is the material homogenous and a true evolution going on, or is the material heterogenous (i.e. a mixture of n homogenous, stable groups)? It was Pearson who took up this issue and provided the mathematical tools to carry out Weldon's analysis.

In the first of his *Contributions to the Mathematical Theory of Evolution* (1894), Pearson devised a moment-based technique for the "dissection of abnormal frequency curves" into mixtures of two normal components. The compounded curve could fit either a symmetric or

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an asymmetric data histogram, depending on whether the mixed distributions had coincident or parallel axes. In the symmetric case—the most relevant to the genesis of kurtosis—the existence of a real solution to the dissection problem was determined by the quantities

$$\varepsilon_1 = \frac{\mu_4 - 3\mu_2^2}{3\mu_2^2}, \tag{2}$$

$$\varepsilon_2 = \frac{\mu_6 - 5\mu_2\mu_4}{5\mu_2^3}, \tag{3}$$

μ_r being the r-th central moment of the observed population.

"I propose to call (ϵ_1) the *excess* and (ϵ_2) the *defect* of the frequency curve. The excess measures the excess of one third of the fourth moment over the square of the second moment; the defect measures the defect of the fourth moment from one-fifth of the ratio of the sixth moment to the second moment" (Pearson, 1894).

Because the normal distribution involved a condition on each pair of even central moments

$$\mu_{2r} = (2r-1)\mu_2\mu_{2r-2} \quad \text{for} \quad r \geq 2, \tag{4}$$

both the excess and the defect would be null if observations came from a Gaussian population. Conversely, non-zero estimates of ε_1 and ε_2 appeared to be "a good measure for practical purposes of the divergence of a given symmetrical frequency curve from the normal type" (Pearson, 1894).

For typical situations in Weldon's data sets, Pearson provided the following interpretations:

1. If the excess and defect were of opposite signs, then the abnormal frequency curve could not be broken up into a mixture of two normal components. In Pearson's view, this pointed to a homogenous but non-Gaussian population, displaying an evolutionary trend in the measured organ.
2. If the excess and defect were both positive, then the measured material was neither homogenous nor Gaussian. It should be modelled, instead, as two superposed normal curves with coincident axes, representing a mixed population of two different, but stable, races. The outcome of such mixture was an excess of frequencies in the neighbourhood of the mean as compared with the corresponding frequencies under the Gaussian model.

In both cases, a positive value of Pearson's excess ($\varepsilon_1 > 0$ or, equivalently, $\mu_4 > 3\mu_2^2$) occurred when the actual distribution was more peaked than the normal curve with the same mean and variance. This relationship between peakedness at the mean and the sign of ε_1 may be seen as the first step in the derivation of a kurtosis measure. As to the defect, Pearson himself recognised the difficulty of getting consistent estimates of the sixth moment and dropped ε_2 from his subsequent research.

3 1895–1901: The Pearson System and Its Characteristic Coordinates

During his work on evolution in cooperation with Weldon, Pearson had repeatedly noticed the inadequacy of the assumptions "confessedly or tacitly involved" in the Gaussian theory of errors:

(α) "Positive and negative errors of the same size are equally frequent";
(β) "There are an indefinitely great number of cause groups associated in producing each individual error";

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(γ) "The contribution towards any individual error of these cause-groups are not correlated among themselves" (Pearson, 1902).

When skewness was detected in Weldon's measurements, axiom (α) was clearly not true; when the excess ε_1 had a sensible value, either (β) or (γ) proved invalid. Hence, "we can only conclude from the investigation here considered that the normal curve possesses no special fitness for describing errors or deviations such as arise in observing practice or in nature. We want a more general theoretical frequency. . ." (Pearson, 1894).

In the second of his *Contributions to the Mathematical Theory of Evolution* (1895), Pearson realised that the probabilities for the hypergeometric distribution were not subject to axioms (α)–(γ). He therefore took these probabilities as his starting point and smoothed them in an attempt to construct continuous density functions. The derivation led to the well-known differential equation

$$\frac{dp(x)}{dx} = \frac{(x-a)p(x)}{b_0 + b_1 x + b_2 x^2}, \tag{5}$$

whose solutions $p(x)$ were originally classified into four types of curves besides the Gaussian (Pearson's Types I, II, III, and IV may be found in Pearson, 1895, pp. 367, 372, 373, and 376, respectively).

Although equation (5) involved a small number of parameters, it was sufficiently general to yield a wide variety of shapes (skewed or symmetric; limited or unlimited range; bell-, U-, or J-shaped; etc.), including the normal as a limiting case. An important advantage of the Pearson system was that the four constants a, b_0, b_1, and b_2 could be expressed in terms of the mean μ and the three central moments μ_2, μ_3, μ_4.

3.1 The Criterion k_1 of Type

According to Pearson (1895), "the first stage in determining the type of curve suitable for a given set of observations" was the calculation of the expression $2\mu_2(3\mu_2^2 - \mu_4) + 3\mu_3^2$, whose sign could be used to ascertain whether a Pearsonian frequency curve had limited range, range limited in one direction only, or unlimited range. Dividing by μ_2^3 and changing signs, this gave the criterion

$$k_1 = 2\beta_2 - 3\beta_1 - 6$$

based on the moment ratios defined in equation (1).

3.2 The Improved Criterion k_2 and the Role of β_2

In the following years, Pearson's attention was drawn to two further types of curves "of considerable interest". Although quite common in both sociological statistics and biological investigations, these curves were not covered by the simple criterion k_1. Therefore, in a supplementary memoir issued in 1901, Pearson proposed "an improved criterion of type"

$$k_2 = \frac{\beta_1(\beta_2+3)^2}{4(4\beta_2 - 3\beta_1)(2\beta_2 - 3\beta_1 - 6)},$$

which allowed the classification of up to six types of curves (Types V and VI were introduced in Pearson, 1901, pp. 446 and 448, respectively).

With the values of k_2 being determined by the two moment ratios β_1 and β_2, it became progressively clear that these "fundamental constants" provided a complete taxonomy of the

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Pearson system. This property inspired the construction of a moment-ratio diagram with β_1 as abscissa and β_2 as ordinate, which a member of the Biometric Laboratory at University College (Rhind, 1909) presented "upside-down" (i.e. with increasing values of β_2 corresponding to downward motion). "By aid of this diagram and a reasonable consideration of the probable errors of his β_1 and β_2 the statistician (could) readily determine within what limits he is justified in using any special type of frequency curve for given data" (Rhind, 1909). In a final revision operated by Pearson (1916), the diagram enabled the classification of 12 different types of frequency curves (cf. figure 6.1 in Stuart & Ord, 1994).

In this so-called $(\beta_1,\ \beta_2)$ chart, the normal curve was displayed as the point of coordinates $\beta_1 = 0,\ \beta_2 = 3$. In addition, several other forms of symmetrical distributions appeared along the vertical axis $(\beta_1 = 0)$ and were classified according to β_2 values. As a "characteristic coordinate" of the Pearson system, β_2 retained the same information as the excess ε_1 and, after 1901, helped Pearson face criticism against his newborn system of nonnormal curves.

4 1902: The Concept of Flat-Toppedness

Pearson (1902) contains a further attack on contemporary attitudes that ignored departures from normality in empirical distributions. Once again, Pearson's criticism focused on the most common features he had encountered during his work on evolution: skewness and non-Gaussian types of symmetry.

As a consequence of axiom (α) in the Gaussian theory of errors (Section 3), fitting an observed frequency distribution with the normal curve necessarily involved a symmetry assumption. However, "in an immense number of cases, such as occur in organic nature, this symmetry is quite unknown" (Pearson, 1902).

Pearson (1895) had already spoken of "skew frequency distributions" when the mode diverged from the mean, suggesting the skewness index

$$\chi = \frac{mean - mode}{\sigma} \tag{6}$$

for use with his system of distributions. Because Pearson's frequency curves were uniquely identified by their first four moments, equation (6) could be specified as a function of the characteristic coordinates $(\beta_1,\ \beta_2)$

$$\chi = \frac{\sqrt{\beta_1}\,(\beta_2 + 3)}{2\,(5\beta_2 - 6\beta_1 - 9)}.$$

It followed that $\beta_1 = 0$ was a sufficient condition for symmetry within his system, but only a necessary one for normality.

As already mentioned in Pearson (1894), the normal distribution involved further conditions (4) as to the even moments, depending on axioms (β) and (γ) of the Gaussian theory of errors. The simplest of such relations, i.e.

$$\mu_4 = 3\mu_2^2 \tag{7}$$

was sufficient to test, within the Pearson system, whether a given symmetrical distribution was a true normal curve.

4.1 An Example of Flat-Toppedness

An example given in Merriman (1891, p. 14) helped Pearson illustrate the influence of equation (7) on distributional shape. Using the details of 1 000 shots at a target provided by

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Belt	Frequency (Observations)	Frequency (Normal curve)
1	1	1
2	4	6
3	10	27
4	89	67
5	190	162
6	212	242
7	204	240
8	193	157
9	79	70
10	16	26
11	2	2

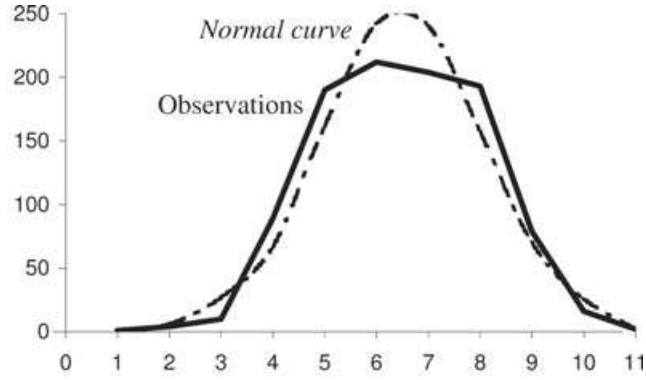


Figure 1. *The distribution of the results of 1 000 shots at a target (Merriman's data set): observed frequencies and the corresponding normal curve. Based on Pearson's (1900) table (p. 173).*

Merriman, Pearson computed the second and fourth central moments

$$\mu_2 = 2{,}4023, \quad \mu_4 = 14{,}5785.$$

Accordingly, he found that

$$\mu_4 - 3\mu_2^2 = -3{,}7353 < 0, \tag{8}$$

and the observations, although nearly symmetrical, were "immensely flatter than the normal curve with which they could be fitted" (actually, Pearson used to scale moments by the sample size, so his value of $\mu_4 - 3\mu_2^2$ corresponds to equation (8) multiplied by a factor of 1 000).

In Pearson's view, the sign of equation (8) was a consequence of the fact that the maximum ordinate of the normal curve exceeded by nearly 20% the corresponding ordinate of the observations (Figure 1). This particular type of departure from normality, which had nothing to do with either skewness or multimodality, was described by Pearson in terms of *flat-toppedness*: "Symmetrical or nearly symmetrical curves which have the same mean and standard deviation as a normal curve will be *flatter topped* if μ_4 be $<3\mu_2^2$, and *steeper at the top* if μ_4 be $> 3\mu_2^2$" (Pearson, 1902).

As the idea embedded in the excess ε_1 was becoming more and more precise, Pearson was getting nearer to his kurtosis coefficient.

5 1905: The Degree of Kurtosis

Although the Pearson system yielded simple mathematical representations for histogram data in many applications, a number of Pearson's contemporaries disregarded the need for generalised frequency curves in either anthropometric or biological statistics. Ranke & Greiner (1904), for instance, criticised an example given in Pearson (1895) in which the measurements of 900 "Altbaierisch crania" had been fitted with a Type IV distribution (Figure 2). Because the data looked approximately symmetric and were "not on the average incompatible with the Gaussian law", the two German scientists found it "idle" to apply a generalised curve of the Pearson system instead of a much simpler normal curve.

Pearson's reply soon appeared in *Biometrika*, the journal he had founded together with Galton and Weldon in 1901. In this rejoinder, usually quoted as the first reference to kurtosis, Pearson (1905) stressed that: "the two tests (a) whether special physical constants (of a distribution) satisfy the Gaussian law, and (b) whether the general distribution of frequency satisfies within

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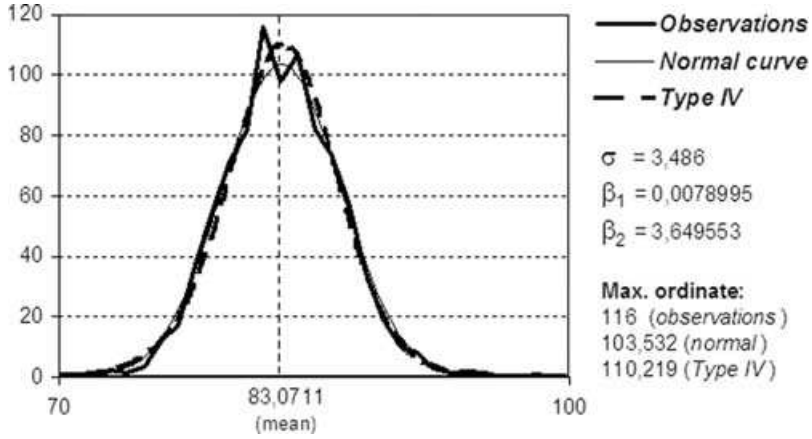


Figure 2. *The distribution of the length–breadth index of 900 Bavarian skulls: observed frequencies, normal curve, and curve of Type IV. Based on Pearson's (1895) table, example V (p. 388).*

reasonable limits the Gaussian law", were not necessarily identical. Although (b) was satisfied for the Bavarian skulls data, the same was not true of (a) because the characteristic coordinate β_2 deviated significantly from its Gaussian value of 3. For the difference $\beta_2 - 3 = 0{,}65$, Pearson obtained a probable error of about 0,11 under the normality assumption, using formula (xxxix) given in Pearson (1902, p. 278). Accordingly, it was "extremely improbable that Ranke's measurements were truly given by a Gaussian distribution in all their features". Conversely, a generalised frequency curve allowing for slight skewness and peakedness at the mean would provide a better solution to the curve-fitting problem.

Pearson's argument used the quantity

$$\eta = \beta_2 - 3 = \frac{\mu_4 - 3\mu_2^2}{\mu_2^2} \tag{9}$$

as "a measure of whether the frequency towards the mean was emphasised more or less than that required by the Gaussian law".

For a better characterisation of this departure from normality, Pearson replaced the previous terms "excess" (1894) and "flat-toppedness" (1902) by the Greek word "kurtosis" (= curvature). Given two symmetrical distributions with equal total frequency and the same standard deviation, the *degree of kurtosis* in equation (9) would qualify them as "relatively more or less flat-topped than the corresponding normal curve". In the paper of 1905, Pearson termed a symmetrical curve *platykurtic* if more flat-topped than the normal, *leptokurtic* if less flat-topped, and *mesokurtic* if equally flat-topped. Within the Pearson system, the conditions for platykurtosis, leptokurtosis, and mesokurtosis were given, respectively, by $\eta < 0$, $\eta > 0$, and $\eta = 0$. Hence "no distribution could be legitimately described as normal or Gaussian" unless both η (the kurtosis) and χ (the skewness) be zero.

5.1 An Example of Leptokurtosis and Platykurtosis

In his attempt to prove that "the mesokurtosis of the Gaussian curve is not a universal characteristic of frequency distributions", Pearson compared two symmetrical, bell-shaped curves of his system with a normal distribution with the same mean and standard deviation.

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The first was a Type II_L curve, having limited range

$$|x| \leq \sigma\sqrt{2\left(m_1 + \frac{1}{2}\right)},$$

and probability density function

$$y = y_1\left[1 - \frac{x^2}{2\sigma^2(m_1 + \frac{1}{2})}\right]^{m_1 - 1},$$

with y_1 as a normalising constant (giving height of the curve at the origin) and

$$m_1 = \frac{3(\beta_2 - 1)}{2(3 - \beta_2)} > 1.$$

The other was a curve of Type *VII* (Student's t), with unlimited range and probability density function

$$y = y_2\left[1 + \frac{x^2}{2\sigma^2\left(m_2 - \frac{1}{2}\right)}\right]^{-(m_2+1)}$$

defined for

$$m_2 = \frac{3(\beta_2 - 1)}{2(\beta_2 - 3)} > \frac{3}{2},$$

and y_2 as a normalising constant with the same role as y_1 in the previous curve (for a detailed characterisation of both curve types and their parameters, the reader is referred to Pearson (1916, pp. 449–450, which also corrects "some rather disturbing misprints" appearing in Pearson (1905)). In the 1916 parameterisation, which has been maintained in subsequent literature, the symbol m replaces both $m_1 - 1$ [Type II_L] and $m_2 + 1$ [Type *VII*].

Because both curves had zero mean and standard deviation σ, "no Gaussian fitting could distinguish one (...) from the other". However, they would only pass into the corresponding Gaussian curve

$$y = y_0 \exp\left[-\frac{1}{2}\left(\frac{x}{\sigma}\right)^2\right], \quad \text{with } y_0 = \frac{1}{\sqrt{2\pi}\sigma}$$

when "m_1 and m_2 were theoretically infinite, practically large". The difference between either distribution and the normal arose from the degree of kurtosis:

$\eta = \frac{-6}{3+2m_1} < 0$ for the first curve;
$\eta = \frac{6}{2m_2-3} > 0$ for the second curve; and
$\eta = 0$ for the normal curve.

(By a slip, Pearson mixed up the signs of η for curves (10) and (11), ending up in a reverse characterisation of leptokurtosis and platykurtosis. As noted by Dyson (1943), the slip was repeated by successive authors, such as Aitken (1939)).

Elaborating on Pearson's discussion, we have formalised the relationship between the values of η and the curvature in the neighbourhood of the origin using a second-order Taylor series expansion. For the first curve, this approach gives

$$\frac{y}{y_1} = 1 - \frac{m_1 - 1}{m_1 + \frac{1}{2}} \cdot \frac{x^2}{2\sigma^2} + o(x^2); \tag{10}$$

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for the second,

$$\frac{y}{y_2} = 1 - \frac{m_2 + 1}{m_2 - \frac{1}{2}} \cdot \frac{x^2}{2\sigma^2} + o(x^2); \tag{11}$$

and for the normal,

$$\frac{y}{y_0} = 1 - \frac{x^2}{2\sigma^2} + o(x^2).$$

Because

$$\frac{m_1 - 1}{m_1 + \frac{1}{2}} < 1 \quad \text{and} \quad \frac{m_2 + 1}{m_2 - \frac{1}{2}} > 1,$$

it is easy to show that

$$\frac{y}{y_2} < \frac{y}{y_0} < \frac{y}{y_1}$$

for values of x close to zero (Figure 3, top). It follows that $y_1 < y_0 < y_2$ in the neighbourhood of the origin (Figure 3, bottom), so "the first curve is flatter-topped than the second and both lie

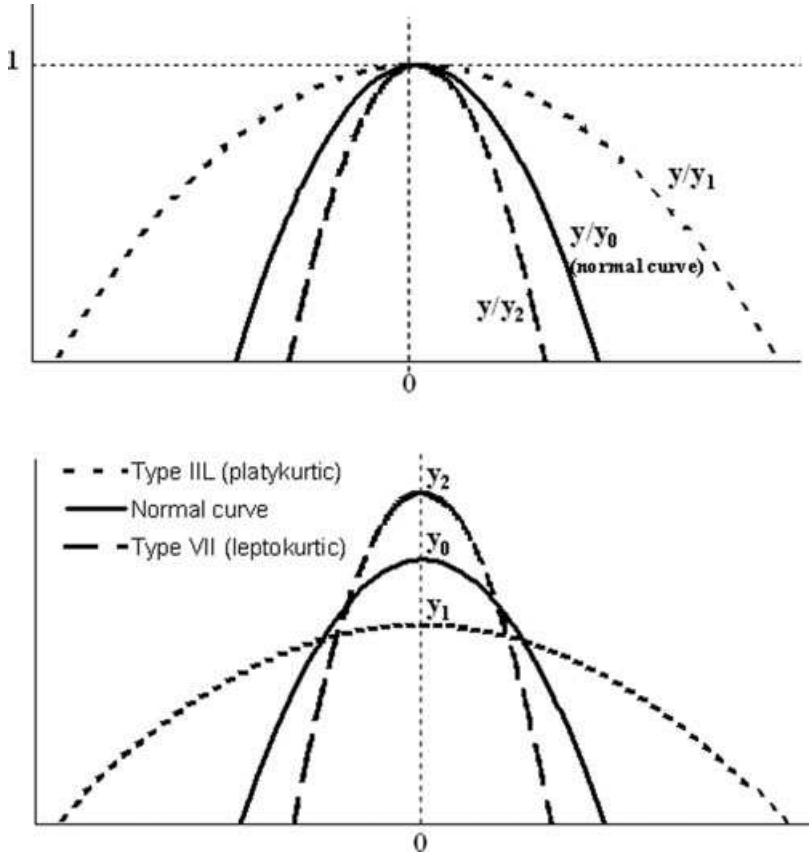


Figure 3. *The qualitative comparison between the normal curve and Pearson Types II_L and VII. Top panel: second-order Taylor series expansions in the neighbourhood of the origin. Bottom panel: the relationship between modal frequencies. Based on Pearson's (1905) example (pp. 174–175).*

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on different sides of the corresponding Gaussian curve. The first curve type is platykurtic and the second leptokurtic" (Pearson, 1905).

This result is a consequence of the fact that, within the Pearson system, $\eta < 0$ implies platykurtosis (first curve), $\eta > 0$ leptokurtosis (second curve), and $\eta = 0$ mesokurtosis (normal curve). Although the degree of kurtosis η embeds the idea of non-Gaussian types of symmetry, the Pearson system provides a method of formalising it in the description of observed frequency distributions.

6 Concluding Remarks

We have described how gradually the idea of kurtosis emerged from Pearson's work on evolution during the period 1894–1905.

At the same time, Pearson was developing his well-known system of frequency curves that displayed a one-to-one correspondence between symmetrical members and β_2 values. As a consequence of this property, the kurtosis coefficient $\eta = \beta_2 - 3$ ordered Pearson's symmetrical curves according to their degree of flat-toppedness (no reference to tail heaviness appeared in Pearson's original derivation).

Outside the Pearson system, however, any combination of peakedness at the mean and kurtosis might occur (Kaplansky, 1945), and the condition $\eta = \chi = 0$ is not sufficient to identify the Gaussian curve (Balanda & MacGillivray, 1988). This partly explains why the treatment of kurtosis in statistical literature has tended to be somewhat fragmented and loose.

As argued by Fiori & Zenga (2005), a proper interpretation of kurtosis in a general context involves both the central and the extreme frequencies. Fung & Seneta (2007) illustrate this point for distributions that arise in financial mathematics in which the concept of kurtosis is resurgently playing a role. For a detailed review of modern approaches to kurtosis measurement, the reader is referred to Zenga (2006). Some recent advances in kurtosis methodology are also discussed in Fiori (2008).

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